



Course Syllabus

Course Title	Advanced Database Systems			
Program	B.Sc. in Software Engineering		Year/Semester: G3SE Sem I	
Module Title	Database Management Systems		Course Code: SEng4131	
			ECTs: 5 ECTs	
Course Chair	DB and Intelligent System		Module Number: 04	
Prerequisite	Fundamentals of Database Systems		Credit Hrs: 3	
Status of Course	Compulsory		Program: Regular	
Contact Hour/Week	Lec. Hrs: 2 /week	Lab Hrs: 3 /week	Tutorial Hrs: 0	Home Study: 5
Instructor	Melaku M.	Email: melakumeskele143@gmail.com		
Course Description	Advanced Database Systems extends students’ knowledge of database management with emphasis on transaction management, concurrency control, query processing and optimization, database security and recovery, distributed and object-oriented databases, data warehousing, and data mining. The course combines theoretical concepts with practical laboratory activities and projects to develop students’ ability to design, implement, optimize, secure, and evaluate modern database systems and data-intensive applications.			
Course Learning Outcomes (CLOs)	At the end of this course the students will be able to: ▶ CLO-1: Analyze and apply transaction management, concurrency control, and database recovery techniques to maintain data consistency and integrity. ▶ CLO-2: Analyze and optimize database queries using query-processing techniques, indexing strategies, execution plans, and optimization methods. ▶ CLO-3: Design and implement database integrity, authentication, authorization, access-control, and data-protection mechanisms. ▶ CLO-4: Design and implement distributed database systems in homogeneous and heterogeneous environments, considering data			

	fragmentation, replication, distributed query processing, transactions, and recovery. ▶ CLO-5: Apply object-oriented and object-relational database concepts to model and manage complex data structures. ▶ CLO-6: Design and implement basic data-warehouse architectures and apply appropriate data-mining techniques to extract useful information from large datasets. ▶ CLO-7: Develop a practical database solution collaboratively and communicate its design, implementation, performance, security, and limitations effectively.
Teaching Methodology	The course employs a student-centered and competency-based approach combining interactive lectures, demonstrations, laboratory exercises, tutorials, problem-based learning, case studies, assignments, group discussions, project work, and student presentations. These methods emphasize practical application, problem-solving, teamwork, and self-directed learning in line with the Outcome-Based Education (OBE) framework.

Mapping CLO to PLO and Assessment Method

CLO	Linked PLOs	Assessment Methods
CLO-1	PLO-1 (Apply knowledge of computing fundamentals) PLO-2 (Identify, formulate, and solve complex engineering problems.) PLO-5	Quiz, Mid-Term Exam, Final Exam
CLO-2	PLO-2 (Identify, formulate, and solve problems) PLO-4 (Analyze and interpret data)	Assignment, Lab Exercise, Mid-Term Exam, Final Exam
CLO-3	PLO-3 (Design solutions) PLO-5 (Apply modern tools and techniques) PLO-7 (Recognize ethical and professional responsibilities)	Quiz, Lab Exercise, Project, Final Exam
CLO-4	PLO-2 (Identify, formulate, and solve complex engineering problems.) PLO-3 (Design solutions) PLO-6 (Analyze impacts of sustainable development)	Assignment, Project, Final Exam
CLO-5	PLO-3 (Design solutions) PLO-5 (Apply modern tools and techniques)	Quiz, Lab Exercise, Project, Final Exam
CLO-6	PLO-5 (Apply modern tools and techniques) PLO-4 (Analyze and interpret data)	Project, Final Exam
CLO-7	PLO-8 (Function on a team) PLO-9 (Communicate effectively) PLO-10 (Apply management principles)	Team Project, Project Presentation, Peer Evaluation

A. Course Content and CLO Mapping and Teaching Methods

Date	Topics and Sub Topics	CLO	Teaching method
Week 1 - 2	Chapter 1: Transaction Management and Concurrency Control 1. Transaction 2. Transaction Support 3. Concurrency Control 4. Problems of Concurrent Sharing 5. Concept of Serializability 6. Concurrency Control Mechanism 7. Database Recovery 8. Transaction and Recovery 9. Recovery techniques and facilities	CLO-1	Interactive Lecture, Demonstration, Lab Exercise, Group Discussion, Quiz
Week 3 - 4	Chapter 2: Query Processing and Optimization 2.1 Overview 2.2 Query Processing steps 2.3 Query Decomposition 2.4 Optimization Process 2.5 Approaches to Query Optimization 2.6 Transformation Rules 2.7 Implementing relational Operators 2.8 Pipelining	CLO-2	Interactive Lecture, Demonstration, Lab Exercise, Problem-Solving, Case Study, Assignment

Week 5 - 7	Chapter 3: Database Integrity, Security and Recovery 3.1 Integrity 3.1.1 Integrity Concept & Subsystem 3.1.2 Integrity Constraints 3.1.3 Types of Constraints 3.2 Security 3.2.1 Database threats 3.2.2 Identification and Authentication 3.2.3 Categories of Control 3.2.4 Implementation of Security Subsystem 3.2.5 Data Encryption	CLO-3	Interactive Lecture, Demonstration, Lab Exercise, Group Discussion, Quiz Case Study,
Week 8	Mid-Semester Examination		
Week 9 - 11	Chapter 4: Distributed Database Systems 4.1 Concepts of Distributed Databases 4.2 Distributed Database Design 4.3 Distributed Query Processing and 4.4 Distributed Transaction Management and Recovery	CLO-4	Interactive Lecture, Demonstration, Lab Exercise, Problem-Based Learning, Group Project, Assignment
Week 12 - 13	Chapter 5: Object Oriented DBMS 5.1 Object Oriented Concepts (Abstraction, Encapsulation, and Information hiding) 5.2 Drawbacks of relational DBMS 5.3 OODBMS definitions 5.4 OO Database Design and Implementation 5.5 OO Data modeling and E-R diagramming 5.6 ObjCP and Attributes 5.7 Object Identity 5.8 Storing objCP in relational database systems Assignment	CLO-5	Interactive Lecture, Demonstration, Lab Exercise, Quiz

Week 14 - 15	Chapter 6. Data warehousing and Data Mining Techniques 6.1 Data Warehousing 6.1.1 Introduction 6.1.2 Benefits 6.1.3 Online Transaction Processing (OLTP) and Data Warehousing 6.2 Data Mining 6.2.1 Introduction 6.2.2 Data Mining Techniques	CLO-6	Interactive Lecture, Demonstration, Lab Exercise, Case Study, Project Work
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B. Assessment Methods

- ▶ Assignment & Quizzes 20%
- ▶ Mid-Exam 20%
- ▶ Project & Lab activity 20%
- ▶ Final examination 40%

C. Course Resources

Textbook

1. Connolly T., Beg C. and Strachan A.: Database Systems, 3rd and 4th ed. Addison-Wesley Publishing Company.
2. RamezElmasri, Shamkant B. Navathe, Fundamentals of Database Systems (4thed), USA, Addison-Wesley, 2004
3. C.J. DATE, Introduction to Database systems (8thed), USA, Addison Wesley, 2003 , Abraham Silberschatz, Henry F. Korth, S. Sudarshan,
4. Database System Concepts (4thed), USA, McGraw Hill Inc., 2002.